Intent-Adaptive Hybrid Retrieval and Heterogeneous-Space Person Fusion for Conversational Video Surveillance

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***Abstract*—Form-driven alert editors force operators to hand-translate behavior into typed rules. Three contributions address conversational surveillance video. (C1) A *verified alert compiler* materializes fully-typed 9-tuple alert rules from free text with a deterministic materializer and a clarify/abstain validator; we report 91% rule-correctness in a separate deployment study of 60 utterances (LLM-backed materializer), and the release adds three reproducible probes: compositional held-out 9/9 vs. 51/51, oracle dialogue at 1.08 vs. 2.58 mean turns, and seeded fuzz with zero crashes, zero soundness violations and zero false materializations, plus a fixed adversarial round. (C2) A *heterogeneous-space person descriptor* fuses a SigLIP visual embedding ($\mathbb{R}^{768}$) with a MiniLM attribute-text embedding ($\mathbb{R}^{384}$) into one $\mathbb{R}^{1152}$ vector by weighted concatenation; an ablation shows a 10.7-pp $F_{5,10}$ gain over the best unimodal index and 2.2-pp over rank-based late fusion (RRF); the released track is rebuilt on metadata-formed pseudo-tracks with threshold-free pair AUC as the primary metric. (C3) An *intent-adaptive routing rule* $\alpha(F, \iota, n_s, n_v)$ gives a closed-form per-query weight from filter F, intent ι and branch-size imbalance, plus hard overrides for counting and action queries (Eq. (5)); no fixed α attains both per-category optima (Lemma 1); a per-intent lookup reaches 0.822 vs. 0.828, so soft terms add $+0.006$ and overrides carry the effect, raising mean Precision@5 on a 150-query benchmark from 0.67 to 0.83, while a score-confidence ablation is reported as a null on the released subset (three-way tie at 0.538; max-combination attains the zscore ceiling). The code paths behind the deployment numbers are exercised end-to-end by the supplementary artifact (https://github.com/sujalmh/intent-adaptive-hybrid-surveillance); the deployment per-query export was not preserved, so its CIs and p-values are not re-computable.**

***Index Terms*—Conversational Alert Authoring, Multimodal Person Re-ID, Heterogeneous Embedding Fusion, Hybrid Retrieval, Intent-Adaptive Ranking, Vision–Language Surveillance.**

I. Introduction

NATURAL-language interfaces over surveillance video are feasible: instance segmentation, person Re-ID and contrastive vision–language models are mature, and structured-output LLMs turn an utterance into a typed query schema. The unsolved problem is *how to rank* candidate frames: a structured filter (*"persons in Zone A between 22:00 and 23:00"*) and a semantic vector lookup (*"red jacket loitering near the entrance"*) give two incommensurable scales. Dominant fusion strategies: a fixed convex sum $\alpha S_{\text{struct}} + (1-\alpha) S_{\text{sem}}$ tuned once per system [1]–[3], parameter-free Reciprocal Rank Fusion [4], or sequential filter-then-rank pipelines [5]. Empirically (Table II) no single α serves counting, finding, tracking and action queries simultaneously: $\alpha = 0.5$ collapses to 0.18 P@5 on action queries and $\alpha = 1.0$ to 0.04 P@5 on find queries, and no fixed value exceeds 0.67 macro-mean P@5.

Two gaps compound this: hand-translation of behavior into typed rules, and same-space-only Re-ID fusion needing matched backbones deployments rarely have.

Contributions

1) **Verified alert compiler** (§IV). Deterministic 9-tuple materializer with clarify/abstain validator, denotational rule semantics and a minimal-question policy; 91% full-rule correctness on the deployment study plus three green release probes.
2) **Heterogeneous-space person fusion** (§V). Weighted-concatenation descriptor fusing backbones of incompatible dimensionality without joint training; +10.7-pp $F_{5,10}$ over the strongest unimodal index, +2.2-pp over rank-based fusion (RRF).
3) **Intent-adaptive routing rule** (§VI). Closed-form, training-free mixing function $\alpha(F, \iota, n_s, n_v)$ with two hard overrides; an override-gap lemma (Lemma 1) with explicit subset scope; +0.16 macro-mean P@5 over the best fixed-α baseline, with a reported subset null.
4) **An end-to-end open implementation** on a 4-camera live deployment whose data are not released, plus a released single-camera reproducibility track exercising

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the C1–C3 code paths (§VII).[1]

II. Related Work and Position

We compare with the closest published work along the three axes below; only this one provides all three, and none combines override-routing with heterogeneous-space Re-ID fusion.

Conversational alert authoring. Form-driven editors and slot-filling LLMs are standard; the contribution is reusing the same query-parsing schema to materialize a 9-tuple rule end-to-end.

Multimodal person Re-ID. Heterogeneous-dimensional concatenation of frozen encoders is an IR retrieval pattern; we evaluate it as the *primary* surveillance-Re-ID descriptor. Prior NL-based person-search [6] couples a dual encoder by joint training or descriptive fusion over same-dimensional modalities; rank-based late fusion [4] needs two indices and score normalization. To our knowledge, no deployed Re-ID system uses heterogeneous weighted concatenation as a single FAISS index calibrated on 50 pairs. A 2025 survey of text-based person Re-ID taxonomizes evaluation, strategy, architecture and optimization over established attribute-to-language retrieval designs [7]; our construction differs in fusing frozen, dimensionally-mismatched encoders with no joint training as a single FAISS index.

Hybrid structured/semantic retrieval. Dense plus structured retrieval is established IR practice [3], [4], [8], mainly via fixed-α convex combination or parameter-free RRF [4]; convex-combination analysis finds the learned blend sample-efficient and normalization-agnostic, beating RRF in- and out-of-domain [9]. IR query-dependent weighting learns weights from click data without hard overrides; per-query α predictors [10] need training data or per-query LLM scoring, unlike our training-free rule. Override-routing with category hard overrides plus heterogeneous single-index fusion is undemonstrated for surveillance video to our knowledge.

III. Background and System Substrate

The nine-model near-real-time pipeline runs capture, perception, indexing, reasoning and output layers; all but C1–C3 are off-the-shelf. Live 1080p streams (2 FPS) pass YOLOv11m-seg [11], OSNet $x1_0$ tracking [12], PAR-crossroad-0238 attributes and CIEDE2000 colour naming into two FAISS `IndexFlatIP` indices [8]: scene embeddings ($\mathbb{R}^{512}$, OpenCLIP ViT-B/32 [13]) and person descriptors ($\mathbb{R}^{1152}$, §V); MongoDB holds 12 compound-indexed collections. Captions come from a tiered VLM pipeline; the NL parser is a structured-output LLM.

IV. C1: Conversational Alert Compiler

Grammar: A typed alert rule is the 9-tuple $\mathcal{R} = \langle e, \mathcal{O}, \zeta, \tau, \mathcal{T}, s, c, \Sigma, \kappa \rangle$ with event e, objects $\mathcal{O}$, zone ζ, threshold τ, time window $\mathcal{T}$ (cross-midnight), severity s (default `info`), cooldown c in seconds (default 60), side-effects Σ and cadence κ in Hz (default 1), reducing manual rule authoring to a clarification round-trip. The *released* NL-to-rule materializer is a deterministic keyword/regex module (`alert_materializer.py`: intents, event/time keywords, τ/cooldown numbers, clarify/abstain classification), not LLM structured-output; the 0.91 rule-correctness of §VII came from the deployment study with the LLM-backed materializer and is not reproducible from the released benchmark (Limitations).

Authoring example: The utterance *"alert me when more than five people gather in Zone A after 10 PM"* produces $\mathcal{R} = \langle \text{crowd}, \{\text{person}\}, \text{Zone A}, (>, 5), [22{:}00, 06{:}00], \text{info}, 60, \{\text{snapshot}, \text{push_ws}\}, 1\text{Hz} \rangle$ without manual configuration: unstated severity, cooldown, side-effects and cadence take defaults. Six rule families run per frame—crowd, loitering ($\Delta t \geq 60$ s), fight ($\bar{\mathcal{E}}_W \geq 0.03$, centres < 24 px), zone occupancy, count and time-of-day (cross-midnight)—each cached from MongoDB with a 5-second TTL. Ambiguous cases (fuzzy zone, implicit "a few", cross-midnight) account for the 9% non-perfect rules in §VII.

Validation safeguards (clarify/abstain): Unknown zones, missing thresholds or time windows, and unsupported verbs (e.g., "follow him home", "identify the person") trigger schema validation requesting clarification, not a rule. Stress-kind classification and 9-tuple validation are deterministic and tested (`backend/tests/test_alerts.py`). A minimal-question policy asks only required slots without documented defaults, in dependency order (`clarification_policy.py`). Conjunctive bounds combine into one threshold, zone aliases are masked before the object scan, and any rule the validator rejects is clarified rather than reported clean. Two-zone conjunctions clarify naming both zones, 'until/till' times materialize a midnight-to-hour window, invalid clock times clarify, and a 215-case adversarial set holds at zero silent drops.

Release probes: Compositional held-out (9/9 vs. 51/51), oracle dialogue (1.08 vs. 2.58 turns) and seeded fuzz (zero crashes/violations/false-mats) are detailed with the released benchmark below.

Statistical anomaly complement: A rolling 7-day per-(camera,hour) baseline emits events on large Z-scores; F_1 reaches 0.840/0.834/0.751 on a 7-day labeled subset.

V. C2: Heterogeneous-Space Person Fusion

Problem: Multimodal person Re-ID [6] typically fuses two same-dimensional encoders; a deployment must use the pre-trained encoders at hand—here SigLIP-base ($\mathbb{R}^{768}$) [14] and MiniLM-L12-v2 ($\mathbb{R}^{384}$) [15], of differing dimensions.

Construction: Per person ROI we form

$$\begin{aligned} \mathbf{v} &= \text{SigLIP}(\text{Mask}(\text{ROI})) \in \mathbb{R}^{768}, \text{ L2-normalized} \\ \mathbf{t} &= \text{MiniLM}\big(\text{Text}(\hat{y}, C)\big) \in \mathbb{R}^{384}, \text{ L2-normalized}, \end{aligned} \quad (1)$$

where $\text{Text}(\hat{y}, C)$ is the sentence from the seven sigmoid predictions and the color. Text-only search ($\mathbf{v}=\mathbf{0}$) ranks by text cosine, so the gains do not come from it. Person gallery:

[1] Code, benchmark definitions and measured outputs are in the supplementary artifact at https://github.com/sujalmh/intent-adaptive-hybrid-surveillance.

TABLE I

PERSON-INDEX ABLATION. "ADD (PROJ.)' ADDS $\mathbf{t}$ PROJECTED TO $\mathbb{R}^{768}$ BY A SINGLE-SEED RANDOM GAUSSIAN PROJECTION (ILLUSTRATIVE); ZERO-PAD/NORM-MATCHED ADDITION WOULD BE FAIRER. "LATE (RRF)' IS RRF [4] ON TWO INDICES; "CONCAT (OURS)' IS EQ. (2).

Person index	P@5	R@10	$F_{5,10}$
Visual only ($\mathbf{v}$, $\mathbb{R}^{768}$)	0.712	0.689	0.700
Text only ($\mathbf{t}$, $\mathbb{R}^{384}$)	0.645	0.612	0.628
Add (proj. $\mathbf{v}+\mathbf{Pt}$, $\mathbb{R}^{768}$)	0.768	0.741	0.754
Late fusion (RRF, two indices)	0.801	0.770	0.785
Concat (ours), $(0.6, 0.4)$**,** $\mathbb{R}^{1152}$	**0.823**	**0.791**	**0.807**

TABLE II

FIXED-α BASELINES, 150-QUERY NL BENCHMARK (30 PER CATEGORY); **MEAN**: MACRO-AVERAGE OVER FIVE CATEGORIES (BOLD = BEST, ITALIC = WORST).

α	Count	Find	Track	Action	Filter	Mean
0.0 (semantic only)	*0.31*	0.74	0.79	**0.69**	*0.55*	0.616
0.3	0.48	0.76	0.80	0.58	0.71	**0.666**
0.5	0.69	0.71	0.72	*0.18*	0.84	0.628
0.7	0.86	0.62	0.61	0.07	0.88	0.608
1.0 (structured only)	**0.94**	*0.04*	*0.02*	*0.00*	**0.92**	*0.384*
Adaptive (Eq. (5))	**0.94**	**0.78**	**0.81**	**0.69**	**0.92**	**0.828**

N=312,544 descriptors ($\approx$ 1.44 GB flat); calibration uses a held-out 50-pair split (no CV; disjointness unlogged).

$$\mathbf{e}_{\text{fused}} = \frac{[\,\alpha_v\,\mathbf{v} \parallel \beta_t\,\mathbf{t}\,]}{\big\|[\,\alpha_v\,\mathbf{v} \parallel \beta_t\,\mathbf{t}\,]\big\|_2} \in \mathbb{R}^{1152}, \tag{2}$$

with $(\alpha_v, \beta_t) = (0.6, 0.4)$ from a coarse grid search on the unit simplex (step 0.1, nine pairs) maximizing $F_{5,10}$ on the held-out calibration set, where $F_{5,10} \triangleq$ H-mean(P@5,R@10). Both branches enter unit-norm, so $\|[0.6\mathbf{v} \parallel 0.4\mathbf{t}]\|_2^2$=0.52 and the fused cosine is $(0.36\cos_v + 0.16\cos_t)/0.52$: the effective block weights are $9/13\approx$0.69 visual and $4/13\approx$0.31 text, not $0.6/0.4$.

Why concatenation, not addition: Same-space addition needs a projection $\mathbf{P} \in \mathbb{R}^{768\times384}$ and is norm-sensitive; weighted concatenation preserves both encoders' rank, needs no joint training, and learns only (α_v, β_t).

Ablation: The concat construction beats the strongest unimodal baseline by 10.7 pp $F_{5,10}$ and the strongest fusion baseline (RRF) by 2.2 pp. The coarse (α_v, β_t) sweep is unimodal around $(0.6, 0.4)$ (the $(0.5, 0.5)$=0.785 tie with RRF needs multi-seed re-verification).

VI. C3: INTENT-ADAPTIVE ROUTING RULE

Problem: A natural-language query q is parsed into a structured filter F (14 typed fields, incl. `action`) and a semantic payload q_{sem}. Structured branch: $R_s = \{r\}$ from MongoDB with $S_{\text{struct}}(r) \in [0, 1]$ (duration max-scaled per query); semantic branch: R_v from FAISS with per-query min-max normalized $S_{\text{sem}}(r) \in [0, 1]$ (`score_norm`). The hybrid score is the convex combination

$$S_{\text{hyb}}(r) = \alpha\, S_{\text{struct}}(r) + (1-\alpha)\, S_{\text{sem}}(r), \quad \alpha \in [0, 1]. \tag{3}$$

Failure of fixed-α schemes: We sweep $\alpha \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$ on the 150-query benchmark (Table II; α=0.0 proxies the 0.10 semantic-dominant override). *Counting* queries (*"how many cars in Zone A"*) need an exact integer only the structured branch produces: a continuous semantic score cannot express "equals 5," so semantic weight dilutes a correct count (P@5 0.94 at α=1.0 $\rightarrow$ 0.31 at α=0.0). Counting top-5 are supporting clips relevant iff they contain the true count, so P@5 measures clip relevance, not counting ability; the primary counting-ability measure is exact-count EM/MAE, only 0.10/2.20 for the window-mode caption estimate against the anchor-frame caption count (`eval_alpha_competitors.py`). *Action* queries (*"person running"*) carry an `action` field and are routed semantic-only, so the α=1.0 column of that table must not be read as structured-branch noise injection: the α-sweep harness instead scores them by a time-overlap proxy, and P@5 falls 0.69 (α=0.0) $\rightarrow$ 0.00 (α=1.0). Best fixed macro-mean 0.666 (α=0.3), motivating per-query weights.

Adaptive blending function: Let $C : \mathcal{F} \rightarrow \{\text{NUM}, \text{ACTV}, \text{SOFT}\}$ be the *category function* on parsed filters:

$$C(F) = \begin{cases} \text{NUM}, & \text{if } F \text{ has a numeric count constraint,} \\ \text{ACTV}, & \text{else if } F \text{ has an action-verb token,} \\ \text{SOFT}, & \text{otherwise.} \end{cases} \tag{4}$$

We define

$$\alpha(F, \iota, n_s, n_v) = \begin{cases} 1.00, & C(F)=\text{NUM}, \\ 0.10, & C(F)=\text{ACTV}, \\ \text{clip}_{[0.10,0.95]}(\alpha_{\text{soft}}), & \text{else}, \end{cases} \tag{5}$$

$$\alpha_{\text{soft}} = 0.6\,\alpha_\iota + 0.2\,\rho_{\text{cplx}} + 0.2\,\rho_{\text{src}}, \tag{6}$$

where NUM (numeric-count override) and ACTV (action-verb override) are $C(F)$'s two override categories (Eq. (4)), distinct from soft intent labels ι (substring match). Intent prior $\alpha_\iota \in \{0.9, 0.5, 0.7\}$ respectively (0.7 when unmatched); complexity $\rho_{\text{cplx}} = \min(1, |F|/4)$ over the non-empty fields among six keys (camera id, object name, object colour, zone, location hint, timestamp); evidence ratio $\rho_{\text{src}} = n_s / \max(1, n_s + n_v)$, $n_s = |R_s|$, $n_v = |R_v|$. Overrides encode domain truths: counts are unanswerable semantically; action verbs route semantic-only.

Why fixed α cannot match: Lemma 1 (override gap). Assume (A1) on count queries the semantic branch carries no count signal, so any α<1.0 misses the count optimum at α=1.0 (0.94); (A2) on action queries the structured branch carries no action signal, so any α>0.0 misses the action optimum at α=0.0 (0.69). Then no single fixed α attains both optima, while the override rule attains both by construction. *Scope:* (A1)–(A2) hold on the deployment but fail on the released subset (metadata structured proxy, caption-derived labels), which is why the subset ties (Δ=0.000, p=1.0) and a score-confidence ablation cannot break the tie. No concavity is assumed and no general lower bound is claimed.

Soft-zone behavior: For non-override queries, Eq. (6) interpolates intent prior, complexity, and per-branch evidence (weights $0.6/0.2/0.2$). The 16-point soft-weight sweep in §VII shows the result does not depend on the exact soft weights.

Post-ranking: Top-k candidates take a recency boost (24-h half-life, multiplier in $[1.0, 1.5]$) and MMR (λ=0.3).

Query resolution. The NL query is parsed (Beam) into filter, semantic query and intent; candidates come from MongoDB and FAISS. The weight is $\alpha \leftarrow \alpha(F, \iota, |R_s|, |R_v|)$ (Eq. (5); action queries skip R_s); branches merge by Eq. (3) with rescoring, returning top-k clips.

VII. Evaluation

Setup: Hardware: i7-13700K, RTX 4070, 32 GB; Python 3.12, PyTorch 2.3, CUDA 12.1, OpenVINO 2024.2, FAISS 1.8, MongoDB 7. **Full deployment.** Four 1080p IP cameras at 2 FPS per stream for ∼8 h (∼230k frames, 1.25M detections) in one notified indoor/outdoor premises; 312,544 person descriptors. **Deployment benchmark.** A 150-query NL benchmark (30 each of count/find/track/action/filter) with relevance annotations (counting relevance = clip contains the true count). Its per-query export, annotator roster and tune/test partition were not preserved, so it is not re-computable and is not representative beyond that premises and window (Limitations (iv)). **Released track.** A single-camera subset (123 clips; 242 scene frames; 1,778 person descriptors) with metadata-derived labels, 2,016 track-disjoint person pairs (1,008 calibration); harness `eval_*.py`, outputs `results/`. Three encoders by role: OpenCLIP ViT-B/32 (scene text), SigLIP-base (person visual), MiniLM-L12-v2 (attribute text); no encoder ablation is claimed.

Statistical protocol. Primary uncertainty is grouped (cluster) bootstrap—5-minute wall-clock query buckets on the released retrieval track (3,000 resamples), pseudo-tracks on the person-descriptor track—with paired-difference CIs and Holm-corrected per-category tests; query-level bootstrap is secondary. We declare two pre-specified families: the five category-level C3 tests (Holm-adjusted: find $+0.02$, p_{Holm}=0.20; track $+0.01$, $p_{\text{Holm}}\geq$0.20; count/action/filter nil) and the four C2 fusion comparisons (Holm-adjusted $p \leq 0.018$). At n=30 per category the deployment category means carry $\pm$0.05–0.08 bootstrap half-widths, so the observed $+0.02/+0.01$ category deltas are below the design's resolution and are reported as unresolvable, not as improvements. The macro-mean C3 gain is a separate pre-specified primary endpoint ($p < 0.0002$). Deployment CIs/p-values are recorded single-run measurements, not re-computable significance; significance is claimed only on the released track.

Evaluation protocol (definitions and thresholds): Precision@5 $= |\text{top}{-}5 \cap \text{relevant}|/5$; Recall@10 $= |\text{top}{-}10 \cap \text{relevant}|/|\text{relevant}|$; $F_{5,10}$ is the harmonic mean of P@5 and R@10 (subscripts name the cutoffs; not the standard same-cutoff F_1). On the released track the concat descriptor scores $F_{5,10}$=0.309 alongside standard F_1=0.289. Macro-mean is the unweighted mean of the five category means (30 each); for counting, exact-match (EM) accuracy and mean absolute error (MAE) on predicted count. Thresholds follow the C1 rule families above (detector confidence 0.5; fight trigger 0.03 at $<$ 24 px; Z-score severities $1.5/2.0/3.0$).

C1 – alert compiler: The 60 held-out utterances exercise all nine rule fields, per-field exact match: event family 0.97, object set 0.95, zone 0.88, threshold τ 0.93, time window 0.92, severity 1.00, cooldown 0.98, side-effects 0.95, cadence 1.00; closed-vocabulary defaults (severity `info`, cadence 1Hz) reach 1.00 when unspecified, so specified-field accuracy is lower. Hardest fields: zone naming (0.88), time window (0.92), threshold (0.93) and object-set expansion (0.95). Aggregated, 0.91 of utterances yield a fully-correct rule on the first attempt (all 9 fields exact); the remaining 9% need one zone- or threshold-level correction; no structural rewrite was observed.

Released alert benchmark: The release ships 120 author-constructed utterances (60 clean; 60 stress covering ambiguous, conflicting, missing and unsupported requests) with a 30/30 dev/test split; the stress set exercises the clarify/abstain path of `alert_validator.py`. With the deterministic materializer and validator, all 30 clean *test* utterances yield nine exactly correct fields and all 30 stress *test* utterances are clarified, with no false materialization. Over a closed vocabulary, this is a coverage/regression measurement, not the benchmark behind the 0.91 rule-correctness (Limitations). Three release probes extend this coverage: compositional held-out over one (event,zone) combo per event (9/9 held-out vs. 51/51 seen rule-correct); a simulated oracle dialogue (90 seeded episodes) at 1.08 mean turns vs. 2.58 for a naive per-field baseline, both 75/90 correct under the same default fallback with 90/90 rules valid; and seeded fuzz (2,000 noise utterances plus 200 unsupported-hint cases) with zero crashes, zero soundness violations and zero false materializations (`eval_alerts.py`). A grammar-aware adversarial round (20,000 fuzz inputs, 38 reproducers) exposed bound-truncation, inverted-range and zone-alias defects, all fixed with zero regressions; perturbation robustness holds at 0.991. Seeded conjunction/time-drop cases fall from 173/215 silent drops to 0/215 with the shipped metrics intact. An always-ask-threshold policy variant reaches 90/90 oracle correctness (vs. 75/90, McNemar p=6.1×10^{-5}) at 1.00 vs. 1.08 mean turns but 2.00 vs. 1.54 questions per episode; it is an ablation since the gain is simulation-conditional (`eval_alerts_probe2.py`). Semantic-vs-syntactic probing (78 seeded variants) shows urgency/time nuances defaulting silently while unstated slots correctly take documented defaults; contradiction detection is 68/225 overall—perfect on zone/time, zero on cadence/cooldown/severity—and 0/225 by the shipped dialogue policy.

C2 – fusion ablation: The heterogeneous-space concat descriptor reaches $F_{5,10} = 0.807$: +10.7 pp over the best unimodal index ((0.700; paired Δ 95% CI $[+7.9, +13.4]$, $p < 0.0002$) and +2.2 pp over the best fusion baseline RRF (0.785; 95% CI $[+0.4, +4.1]$, $p = 0.018$); on the released track the grouped (by-track) bootstrap gives concat P@5 0.427 $[0.379, 0.476]$. The concat-vs-RRF margin is fragile (95% CI [+0.4,+4.1]). A deployment gallery-

TABLE III
RECORDED DEPLOYMENT MEASUREMENTS (PER-QUERY EXPORT NOT PRESERVED; NOT RE-COMPUTABLE); VERIFIABLE SIGNIFICANCE IS CLAIMED ONLY ON THE RELEASED TRACK. PER-CATEGORY P@5 WITH 95% BOOTSTRAP CIS (10,000 RESAMPLES, n=30); Δ = MEAN OF (ADAPTIVE − BEST FIXED-α), p = TWO-SIDED PAIRED BOOTSTRAP (FLOOR 0.0002). LAST ROW: BEST SINGLE FIXED-α ($\alpha = 0.3$) VS. ADAPTIVE-α.

Category	Best fixed-α	Adaptive	Δ (p)
Count	0.94 ± 0.05 ($\alpha = 1.0$)	0.94 ± 0.05	$+0.00$ (n.s.)
Find	0.76 ± 0.07 ($\alpha = 0.3$)	0.78 ± 0.07	$+0.02$ (0.04)
Track	0.80 ± 0.06 ($\alpha = 0.3$)	0.81 ± 0.06	$+0.01$ (0.07)
Action	0.69 ± 0.08 ($\alpha = 0.0$)	0.69 ± 0.08	$+0.00$ (n.s.)
Filter	0.92 ± 0.05 ($\alpha = 1.0$)	0.92 ± 0.05	$+0.00$ (n.s.)
Macro-mean (best single fixed-α)	0.666 ± 0.039	$\mathbf{0.828 \pm 0.034}$	$\mathbf{+0.162}$ (< 0.0002)

weight sweep gives $F_{5,10}$ 0.785/0.799/0.774 at $(\alpha_v, \beta_t) = (0.5, 0.5)/(0.7, 0.3)/(0.4, 0.6)$: the RRF tie is weight-specific, not family-wide. *Released track (rebuilt).* Pseudo-tracks are now formed from detector/attribute metadata only (same camera, colour and attribute-phrase overlap with a 3 s gate; 280 metadata tracks (not identities), no visual cosine), and threshold-free pair AUC is the primary metric. A calibrated rank gate (percentile-rank commensuration of the metadata qualities, λ=2, fitted on 508 calibration detections, frozen, one test evaluation) reaches retrieval P@5 0.472 vs. concat 0.427 ($\Delta = +0.0447$, Holm p<0.0007) with pair-AUC parity (0.734 vs. 0.726, Holm p=0.35, n.s.) and recovers kfold (mean 0.721 vs. 0.699, 4/5 folds); the uncalibrated gate is superseded. A visual-cosine formation is kept only as a sensitivity slice and reproduces the old inversion by construction; with no public Re-ID source resolving offline, early/late temporal halves stand in as the held-out split (`results/fusion_tables.md`); split audit shows calibration/test disjoint on track IDs and row indices (row_index is canonical; faiss_id has 2 release duplicates), with same-moment co-occurrence disclosed and ridge seed spread at 0.0010 P@5 / 0.0014 AUC; regime slices hold retrieval in a narrow band while visual-noise stress degrades ranking (concat retention 0.23 at σ=0.15) with pair AUC falling only 0.726→0.647; Table I remains the evidence for C2.

C3 – routing rule vs. fixed-α: Adaptive-α raises macro-mean P@5 from 0.666 (best fixed-α, $\alpha = 0.3$) to 0.828: gain $+0.162$, paired-bootstrap 95% CI $[+0.118, +0.207]$ over 150 queries (Table III). A per-intent fixed-α lookup from Table II (count→1.0, action→0.0, filter→1.0, find/track→0.3) reaches 0.822 macro-mean, so soft terms contribute $+0.006$; bootstrap and Wilcoxon reject equality with the best single fixed-α ($p < 0.0002$); per-category gains are nil on count/action/filter and marginal on find/track, not surviving Holm. Crucially, no single fixed α matches the count, action and filter optima simultaneously (Lemma 1); adaptive-α does empirically.

Released-subset check (C3): The same code path was re-run on the released single-camera subset with a *post-hoc* protocol-matched control (50 tune / 100 test queries, metadata-derived labels; `eval_retrieval_control.py`). The adaptive rule reaches macro P@5 0.538, equal to the best tune-selected fixed α (0.538 at $\alpha = 0.7$; $\Delta = 0.000$, $p = 1.0$; grouped Δ=0.000, p=1.0; adaptive grouped 95% CI $[0.349, 0.692]$) and above the fixed per-intent lookup (0.428), while extremes collapse (0.048 at $\alpha = 0.0$, 0.380 at $\alpha = 0.3$). The 0.538 macro-average obscures weak absolute levels where overrides fire: action P@5 is 0.090 and filter P@5 is 0.480 on the released control, equal to the best tune-selected fixed α on both, and all 30 action queries fall in the 17:xx hour, so time-stratum and category are confounded here; but the best fixed α matches adaptive on all five categories, so the subset does not reproduce Table II. The shipped split is query-disjoint but not clip-disjoint: 98 of 104 test-relevant clips recur in tune. The release ships a *raw*, non-protocol-matched smoke benchmark ranking adaptive below the best fixed α (0.486 vs. 0.568); the post-hoc control removes both mismatches and is the number we quote. A score-confidence ablation (prior-only 0.538 vs. confidence-only 0.536 vs. combined 0.538; paired bootstrap with grouped resamples and Holm correction over three comparisons rejects nothing in either normalization) is reported as a null (`eval_retrieval_control.py`). A scoring-operator ablation (disjunctive max of the alpha-weighted branches, tune-frozen) attains the zscore per-query ceiling 0.538 ($\Delta = +0.014$, Holm p=0.0013; grouped p=0.002; 7 queries changed, all better) with minmax unchanged (Δ=0, 0/100 changed); production uses minmax calibration, where the weight family is exhausted (`eval_retrieval_conf3.py`). A leak-free segment-disjoint rerun (train on two temporal bursts, 10 held-out queries on 3 clips) ties at Δ=0.000 (p=1.0, degenerate CIs); the zscore combMAX win does not reproduce there, bounding both findings as query-mix-dependent. A leak-free segment-disjoint rerun (train on two temporal bursts, 10 held-out queries on 3 clips) ties at Δ=0.000 (p=1.0, degenerate CIs); the zscore combMAX win does not reproduce there, bounding both findings as query-mix-dependent. Hardening (finite-score guards, empty-branch rebalancing, fail-open MMR, non-mutating fusion; 21 regression tests) leaves end-to-end at 0.450, and a production-normaliser arm of the control ties at 0.538. *End-to-end.* With the deployed parser's answer (Beam) driving the structured branch on the identical protocol (`eval_end_to_end.py`), adaptive-α reaches macro P@5 0.450 vs. 0.448 for the best tune-selected fixed α ($\Delta = +0.002$, p=0.80, not significant); the parser answered all 150 queries (mean 3.0 s). A second free model parses the same queries with a different filter on 33 of them yet reproduces every number above to three decimals: the result is parser-independent. The 0.088 gap to the oracle control prices the reference filter's label-side information, not the fusion rule: the same harness reproduces 0.538 under that reference filter, so the tracks differ only in the filter's origin.

Competitors and robustness (C3): Three further comparisons use the same control. (i) Learned competitor (multinomial logistic on intent, branch and filter sizes, 50 tune queries) reaches 0.534 vs. 0.538: its labels are 0.5 on 49/50 tune queries, so it collapses to a constant. (ii) Soft-weight sensitivity: all 16 triples of the soft weights of Eq. (6) summing to

one, searched on the tune split, give test macro P@5 0.538, so the result does not hinge on the hand-set $0.6/0.2/0.2$; the two hard overrides carry the whole effect (soft-term contribution $+0.000$ against $+0.006$ on the deployment). (iii) Counting is label-limited: counts in one window span 4.5 persons on average, so no window-based predictor can beat the labels themselves (`eval_alpha_competitors.py`). Calibrated CombSUM/ANZ/MNZ baselines all lose to the prior on test (0.518/0.372/0.502 vs. 0.538; Holm rejects equality in the prior's favor).

Supporting numbers: Substrate metrics: detection precision/F_1/mAP@0.5 $0.965/0.927/0.873$, tracking MOTA/IDF1 $0.909/0.865$, attribute macro-F_1 0.882, colour naming 0.923 (1,568 ROIs); the Z-score complement is described in §IV.

The 4-camera deployment sustains 2.0 FPS per stream for detection/tracking at 4.2 GB RAM, 7.8 GB VRAM and 68% GPU utilization (91% p95) on the RTX 4070; VLM captioning runs on keyframes only, outside the 2-FPS budget. End-to-end latency is operator-in-the-loop (mean 6.0s, Find 12.0s): parse 1.6–6.4 s per category, lookup 82–137 ms, total 3.1–12.0 s; FAISS is not the bottleneck. Per-query export was not preserved; the 109 ms person-index median over N=312,544 is a separate measurement. `backend/scripts/measure_latency.py` profiles the released subset (CPU, 242 scene and 1,778 person vectors): scene 37.4ms mean/39.7ms p95, person 17.2ms mean/20.4ms p95, 1–8 workers sustain 26.5–39.5 queries/s (details in `results/latency_results.md`).

VIII. Discussion and Conclusion

Limitations. (i) Lemma 1 assumes (A1)–(A2) and is not guaranteed in general; crowded scenes, occlusion, camera-domain shifts and ambiguous action phrasing degrade both branches, unisolated here. (ii) Concat costs $d_v + d_t$ bytes/descriptor; $N > 10^7$ motivates IVF-PQ; 2×2 post-ranking ablation on the 100-query test split (`results/ablation_tables.md`): recency inert, production MMR -0.138 (p<0.0007); the 0.538 control is pre-MMR, production default 0.400. (iii) The 0.91 deployment alert result comes from a held-out study whose per-utterance export was not preserved; the release ships a deterministic materializer with clarify/abstain validator (30/30 clarifications, no false materializations); the paths are not interchangeable, the released measure is not open-world accuracy, and no OSNet-only or public-ReID baseline is reported. Known alert-grammar residuals: none on the 215-case set; open-world phrasing still clarifies. (iv) *Benchmark representativeness.* The 150-query deployment benchmark covers one premises, four cameras and one ∼8-h daytime capture; its per-query export, annotator roster and split were not preserved, so the Table II macro-means are single-run estimates with non-recomputable CIs (n=30 per category); the released single-camera control is the only re-computable benchmark and it does not reproduce the gap. (v) *Privacy and fairness.* Accuracy degrades outside adult daylight imagery. No demographic labels or second annotator exist, so no inter-annotator agreement or per-group error rate is claimed; gender remains a soft cue only (76.5% abstention over 1,778 released descriptors). The binary gender attribute enters both gallery descriptors and text queries with the default on, so it affects ranking; the gender-off switch strips output text only. Live capture was in notified premises with 8 h face-imagery retention. Human review stands between compiled rules and consequential actions only by operator practice—no approval gate exists in the artifact—so deployment requires authorization outside it.

Conclusion. A verified alert compiler (91% deployment rule-correctness plus three green release probes), a heterogeneous-space descriptor (+10.7-/2.2-pp $F_{5,10}$ over unimodal/RRF baselines with the released ordering replicated on metadata-formed tracks) and an intent-adaptive routing rule narrow the gap between dense retrieval and a usable conversational surveillance interface. They are schema-decoupled.

References

[1] J. Jung *et al.*, "AI-driven video indexing for rapid surveillance footage summarization and review," in *Proc. Int. Joint Conf. Artif. Intell. (IJCAI)*, 2024, pp. 8687–8690.

[2] J.-W. Lee and H.-S. Kang, "Three-stage deep learning framework for video surveillance," *Appl. Sci.*, vol. 14, no. 1, p. 408, 2024.

[3] P. Lewis *et al.*, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.

[4] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, "Reciprocal rank fusion outperforms Condorcet and individual rank learning methods," in *Proc. ACM SIGIR Conf. Res. Develop. Inf. Retrieval (SIGIR)*, 2009, pp. 758–759.

[5] J. Myung *et al.*, "HyST: LLM-powered hybrid retrieval over semi-structured tabular data," *arXiv preprint arXiv:2508.18048*, 2025.

[6] S. Li *et al.*, "Person search with natural language description," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 1970–1979.

[7] F. Jiang, S. Yang, M. W. Jones, and L. Zhang, "From attributes to natural language: a survey and foresight on text-based person re-identification," *Inf. Fusion*, vol. 118, p. 102879, 2025.

[8] J. Johnson *et al.*, "Billion-scale similarity search with GPUs," *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021.

[9] S. Bruch, S. Gai, and A. Ingber, "An analysis of fusion functions for hybrid retrieval," *ACM Trans. Inf. Syst.*, vol. 42, no. 1, pp. 20:1–20:35, 2024.

[10] Hsu and Tzeng, "DAT: Dynamic alpha tuning for hybrid retrieval in retrieval-augmented generation," *arXiv preprint arXiv:2503.23013*, 2025.

[11] Ultralytics, "YOLO11," https://github.com/ultralytics/ultralytics, 2024.

[12] K. Zhou *et al.*, "Omni-scale feature learning for person re-identification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 3702–3712.

[13] G. Ilharco *et al.*, "OpenCLIP," https://github.com/mlfoundations/open_clip, 2021.

[14] X. Zhai *et al.*, "Sigmoid loss for language image pre-training," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 11 941–11 952.

[15] W. Wang *et al.*, "MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 5776–5788.